Keywoong Bae

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https://kwoongbae.github.io

Introduction

I received my **M.S. from Pohang University of Science and Technology (POSTECH)** in August 2025. Since September 2025, I have been serving in South Korea as **Professional Research Personnel (Alternative Military Service)**, with an expected completion in 2027.

I am highly motivated to analyze **financial phenomena through data-driven research**. Throughout my undergraduate and graduate curriculum, I have utilized a wide range of approaches from classical statistical techniques to state-of-the-art AI. With these experiences, I'm eager to study financial phenomena from two complementary angles: **(i) an engineering viewpoint** focused on improving analytical performance and **(ii) an economic perspective** aimed at deriving practical, real-world implications. With this background and these research interests, I aim to make meaningful contributions to the field.

* This CV is updated on 23, Aug, 2025.

Research Interests

Financial Domain; Data Science; Statistical Modeling; Generative AI; Deep Learning.

Education

- **POSTECH** — Pohang, South Korea
  M.S. in Dept. of Industrial and Management Engineering — *Sep. 2023 - Aug. 2025*
  - Industrial AI program
  - **Thesis Title:** Systemic Cyber Risks and Insurance Regulatory Capital.

- **Inha University** — Incheon, South Korea
  B.S. in Dept. of Industrial Engineering — *Mar. 2019 - Aug. 2023*

Experience

- **Sixty Hertz** — Seoul, South Korea
  Data Scientist (as a Technical Research Personnel) — *Jul. 2025 - **Current***
  - Performing **anomaly detection** using predictions of renewable energy yield from a **VPP**(Virtual Power Plant).
  - **Skill used:** Python, SQL.
  - **Communication Tool:** Slack / Jira, Confluence.

- **AIRM Lab., POSTECH** — Pohang, South Korea
  Graduate Researcher (Advisor: Kwangmin Jung) — *Sep. 2023 - Jul. 2025*
  - Conducted scenario-based analyses on how **systemic cyber risk** affects **insurance regulatory models**.
  - **Teaching Assistant (TA):** Financial Accounting (IMEN203), Spring 2024; Spring 2025.
  - **Skill used:** Python, R.
  - **Communication Tool:** Slack.

- **Informatics and Deep Learning Lab., Inha Univ.** — Incheon, South Korea
  Research Intern (Advisor: Wookey Lee) — *Apr. 2021 - Jun. 2023*
  - Analyzed performance degradation in **Deep Learning** (e.g., Multi-modal and Diffusion generative models) under **adversarial attacks** on input distributions.
  - **Skill used:** Python / Keras, TensorFlow.

- **SL Solution Co. Ltd.** Seoul, South Korea
  Internship course at Software Team *Aug. 2020*
  - Developed a **responsive web application** visualizing COVID-19 patient movement paths on interactive maps by integrating public datasets with **geocoding** through the **Tmap REST API**.
  - **Skill used:** Java, Javascript / SpringBoot.

- **UDMTEK Co. Ltd.** Suwon, South Korea
  Internship course at MES Team *Dec. 2019 – Feb. 2020*
  - Managed database on **MES**(Manufacturing Execution System) and developed a **MVC**(Model, View, and Controller) service for maintenance.
  - **Skill used:** Java, TypeScript / Angular, SpringBoot.

Publications

A. Working Papers

[1] **Bae, K.**, Jung, K., and Zhang, L. (2025). Systemic cyber risks and insurance regulatory capital. Working Paper.

B. International Conferences

[1] **Bae, K.**, Jung, K., and Zhang, L. (2025, Aug.). Systemic Cyber Risks and Insurance Regulatory Capital. *In the 5th World Risk and Insurance Economics Congress (**WRIEC**).*

[2] **Bae, K.**, Lee, S., and Lee, W. (2023, Dec.). Diffusion-C: Unveiling the Generative Challenges of Diffusion Models through Corrupted Data. *In the 37th Annual Conference on Neural Information Processing Systems (**NeurIPS**), Workshop on Diffusion Models.*

[3] Lee, W., **Bae, K.**, Toshpulatov, M., Lee, S. (2022, Jul.). Bimodal Classification with CNN and Sequence Models. *In the 18th Int. Conference on Data Science **(ICDATA)**.*

[4] **Bae, K.**, Lee, S., and Lee, W. (2022, Jan.). Transformer Networks for Trajectory Cl assification. *In the 2022 IEEE International Conference on Big Data and Smart Computing (**BigComp**).*

C. Domestic Journals

[1] Jung, K. and **Bae, K.** (2025). A Classification and Statistical Analysis on Systemic Cyber Risks. *Korean Journal of Insurance (**KJI**)*, 142, p.1-34.

D. Domestic Conferences

[1] **Bae, K.**, Jung, K., Zhang, L. (2025, Aug.). Systemic cyber risks and insurance regulatory capital. *In the Korean Insurance Academic Society **(KIAS)**.*

[2] Jung, K. and **Bae, K.** (2025, Feb.). A Classification and Statistical Analysis on Systemic Cyber Risks. *In the 3rd Social Science Korea **(SSK)** Networking Symposium.*

[3] Jung, K. and **Bae, K.** (2025, Feb.). A Classification and Statistical Analysis on Systemic Cyber Risks. *In the Korean Insurance Academic Society **(KIAS)**.*

[4] **Bae, K.**, Lee, S., and Lee, W. (2022, Dec.). Robustness Analysis of Diffusion Generation model Using Noises and Corruptions. *In the Korean Software Congress **(KSC)**.*

[5] **Bae, K.**, Lee, S., and Lee, W. (2022, Jul.). Robust Multimodal Classification Model Using Homogeneous Features. *In the Korean Computer Congress **(KCC)**.*

Projects

- **Assessment of the Corporate Customer Service and Proposal for the Improvement.**
  Korean Fire Protection Association (KFPA) *Sep. 2024 - Nov. 2024*
  - **Keywords:** Optimization, Linear Programming.
  - Analyzed the attributes of KFPA inspection staff based on employment types (Subsidiary, Contract worker, Outsourcing) and proposed workforce management strategies for enhancing inspection efficiency.
  - Optimized the number of personnel for building inspections using Linear Programming.

- **Data-driven Evaluation for Safety Assessment of the KFPA and its Future Strategy.**
  Korean Fire Protection Association (KFPA) *Apr. 2024 - Jul. 2024*
  - **Keywords:** Indexing, Random Forest, XAI.
  - Conducted data-driven analysis to estimate the efficiency of digital transformation strategies in building inspections.
  - Utilized a machine learning algorithm, such as random forest, to examine the impact of building features on inspection time and employed eXplainable AI (XAI) techniques for ensuring the interpretability of the results.

- **Responsive Web Application Development.**
  SL Solution *Aug. 2020*
  - **Keywords:** REST API, Geocoding, Responsive web service.
  - Developed a responsive web application to store and visualize the location data of COVID-19 confirmed cases. [Github]
  - Used the web development tools, including Spring Boot, MariaDB, and T-Map API.

- **MES Web Application Maintenance Project.**
  UDMTEK *Dec. 2019 - Feb. 2020*
  - **Keywords:** MES service, Responsive web service.
  - Developed a web application for Manufacturing Execution System (MES) services.
  - Implemented the MVC pattern using web development frameworks (e.g., Angular, Spring Boot, and PostgreSQL).

Skills

- **Language:** Korean (Native), English (Advanced).
- **Programming:** Python, R, C++ (Advanced), Java, JavaScript, SQL (Intermediate).
- **AI Framework:** TensorFlow, PyTorch, Scikit-Learn (Advanced).